# TCF-Net: Tri-Modal Credibility Fusion with Selective Abstention for Multilingual and Code-Mixed Fake News and Hate Speech Detection


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***Abstract*—Misinformation on Indian social media is frequently written in code-mixed form, with Tamil or Hindi typed in Roman script alongside English, and it often carries hateful language. Existing detectors are largely monolingual, rely on a single view of the text, and are obliged to label every input. We present TCF-Net, a tri-modal credibility fusion network that combines (i) contextual semantics from the MuRIL encoder with script-agnostic character n-grams, (ii) a 22-dimensional multilingual deception-stylometry vector, and (iii) an 8-dimensional propagation prior obtained from a text-driven Susceptible–Infected–Recovered simulation. A per-sample softmax gate fuses the streams, and a shared trunk drives a veracity head, a SelectiveNet selection head that can abstain, and a jointly trained hate-speech head. On a 5,000-claim benchmark spanning English, Tamil, Tanglish, Hinglish and Hindi, evaluated with a grouped split that prevents duplicate leakage, TCF-Net attains 92.29% accuracy at full coverage, which is the ceiling imposed by deliberately unverifiable claims, and 100.00% accuracy on the 84.7% of claims it chooses to answer under a validation-calibrated, risk-targeted threshold. Hate speech is detected with 96.50% accuracy. The model abstains on every borderline claim while answering all clear ones, illustrating how selective prediction can route uncertain content to human fact-checkers.**



***Index Terms*—fake news detection, code-mixed text, selective prediction, stylometry, multi-task learning, hate speech, MuRIL.**


## I. INTRODUCTION

Online platforms let a rumour reach thousands of readers within minutes, and false stories have been shown to spread farther and faster than true ones because they are novel and emotionally charged [1]. In India the problem is compounded by linguistic diversity: a single post may switch between English and Tamil or Hindi mid-sentence and may use native script or Roman transliteration. Automatic credibility tools trained on monolingual English news transfer poorly to such text.

Three further gaps motivate this work. First, most detectors observe only what a text says, ignoring how it says it, although deceptive writing tends to hedge, exaggerate and urge immediate sharing. Second, propagation signals are informative [2], yet sharing graphs are rarely available when a post first appears. Third, deployed detectors are forced to label every input, producing confident errors on claims that cannot be verified from the text alone, which is precisely where fact-checkers want to be consulted.

We address these gaps with TCF-Net, whose contributions are: (1) a gated fusion of contextual semantics, multilingual deception stylometry and a label-free propagation prior; (2) a SelectiveNet [3] abstention head trained jointly with a hate-speech head; (3) a leak-free evaluation protocol with grouped splits and validation-only, risk-targeted threshold calibration; and (4) an interactive explainable interface.

## II. RELATED WORK

### *A. Fake News Detection for Indian Languages*

Praseed *et al.* [4] fine-tuned XLM-RoBERTa, multilingual BERT and ELECTRA on Hindi news and ensembled them, improving robustness at the cost of running three transformers at inference. Raja *et al.* [5] used MuRIL with gradient accumulation for Dravidian fake news and released the DFND corpus covering Tamil, Telugu, Kannada and Malayalam. Both operate on native-script news and always emit a label.

### *B. Hate Speech in Code-Mixed Text*

Sreelakshmi *et al.* [6] combined multilingual transformer embeddings with cost-sensitive classical classifiers for offensive language in Dravidian code-mixed posts, reporting accuracies in the seventies on Tamil sets, which shows how hard the problem remains. Shared tasks such as HASOC [7] and DravidianCodeMix [8] supply benchmark data, but hate and veracity are generally modelled separately.

### *C. Context, Modalities and Large Language Models*

Phan *et al.* [2] survey graph neural network detectors that exploit news–user propagation graphs, and Tufchi *et al.* [9] review multimodal fusion of text and images; both conclude that complementary evidence helps but rely on data that are often unavailable at posting time. Hu *et al.* [10] find that GPT-3.5 alone trails a fine-tuned BERT detector and propose distilling LLM rationales into a small model. None of these systems offers a reject option; selective classification [3] has, to our knowledge, not been combined with multilingual credibility assessment.

## III. PROPOSED METHOD

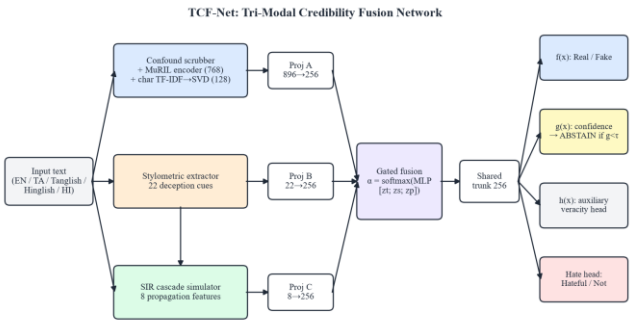



Fig. 1. Architecture of TCF-Net. Three evidence streams are projected to 256 dimensions, fused by a per-sample gate and passed to veracity, selection, auxiliary and hate-speech heads.

### A. Modality A: Contextual Semantics

Input text is normalised and stripped of publisher artefacts (agency datelines, URLs, handles). It is encoded by MuRIL [11], pre-trained on 17 Indian languages and their transliterations; masked mean pooling yields $e \in \mathbb{R}^{768}$. A character 2–5-gram TF-IDF vector, robust to Romanised spelling variation and Indic combining marks, is reduced by truncated SVD to $l \in \mathbb{R}^{128}$. The semantic input is $x_t = [e; l]$.

### B. Modality B: Deception Stylometry

Twenty-two features are computed from the raw text: length statistics; exclamation, question, capitalisation, digit, punctuation and quotation ratios; densities of hedges, sensational terms, superlatives, negations, first- and second-person pronouns and modals; type–token ratio; Flesch ease; affect polarity; and punctuation burstiness. Lexicons cover English, Hindi, Hinglish, Tamil and Tanglish (e.g., *shayad*, *khulasa*, *oruvela*, *adhirchi*).

### C. Modality C: SIR Propagation Prior

An arousal score $a = \text{clip}(4\cdot\text{excl} + 3\cdot\text{caps} + 6\cdot\text{sens}, 0, 1)$ sets the transmission and recovery rates of a discrete-time SIR model [12]:

$$\beta = 0.18 + 0.45a, \quad \gamma = 0.22 - 0.08a \tag{1}$$

$$I(k+1) = I(k) + \beta S(k) I(k) - \gamma I(k) \tag{2}$$

After 24 steps, eight descriptors (peak, time to peak, slope, breadth, depth, echo-chamber, bot-likeness and decay) form $x_p$. The simulator never observes labels, so it cannot leak ground truth.

### D. Gated Fusion

Each modality is projected (Linear–LayerNorm–GELU–Dropout) to $z_t, z_s, z_p \in \mathbb{R}^{256}$. A gate computes

$$\alpha = \text{softmax}(W_2\, \text{ReLU}(W_1[z_t; z_s; z_p])) \tag{3}$$

and the fused vector $z = \Sigma\, \alpha_m\, z_m$ feeds a shared trunk $u$. An entropy bonus on $\alpha$ prevents collapse onto one stream.

### E. Selective Multi-Task Objective

Following SelectiveNet [3], heads $f(u)$, $g(u) \in [0, 1]$ and $h(u)$ produce the prediction, selection score and auxiliary prediction. With batch coverage $\varphi$ = mean $g(x)$:

$$L_{sel} = \Sigma\, g(x_i)\cdot CE(f(x_i), y_i) \,/\, \Sigma\, g(x_i) + \lambda\cdot\max(0, c - \varphi)^2 \tag{4}$$

$$L = \tfrac{1}{2}L_{sel} + \tfrac{1}{2}CE(h, y) - 0.02\cdot H(\alpha) + 0.5\cdot L_{hate} \tag{5}$$

where $c = 0.85$, $\lambda = 32$ and $L_{hate}$ combines main and auxiliary hate-head cross-entropies over labelled samples.

### F. Risk-Targeted Threshold Calibration

At inference the model answers when $g(x) \geq \tau$ and otherwise returns "Not Sure". Instead of a coverage quantile, which shifts with the share of ambiguous claims in each split, $\tau$ is the lowest validation score at which the selective error of answered claims is at most 1%, placed midway to the next lower score ($\tau = 0.375$). The test set is never used for calibration.

## IV. EXPERIMENTAL SETUP

### A. Dataset

We use a 5,000-claim multi-task benchmark generated from templates in five settings: English (1,754), Tamil (949), Tanglish (1,035), Hinglish (746) and Hindi (516). Each claim has veracity, hate-speech and ambiguity labels. 788 claims (15.8%) are *borderline* rumours whose veracity label is assigned at random by design; they bound full-coverage accuracy at roughly 92% and test whether the model learns to abstain.

### B. Leak-Free Protocol

The corpus contains only 1,531 distinct texts. We therefore group samples by normalised text and apply StratifiedGroupKFold (stratified by veracity and language), giving 3,371 training, 630 validation and 999 test claims with no shared text. TF-IDF, SVD and scalers are fitted on training data only.

### C. Implementation and Baselines

TCF-Net is implemented in PyTorch with a frozen MuRIL encoder (128 tokens), AdamW (lr $10^{-3}$), batch 64, cosine annealing and early stopping on validation accuracy (13 epochs, CPU). Baselines use the same split: TF-IDF with Naive Bayes, logistic regression and a linear SVM; stylometric logistic regression; and MuRIL embeddings with logistic regression. Ablations remove Modality C and then Modalities B and C.

## V. RESULTS AND DISCUSSION

### A. Overall Performance

TABLE I
TEST-SET PERFORMANCE OF TCF-NET

| Metric | Value |
|---|---|
| Accuracy (100% coverage) | 92.29% |
| Macro-F1 / ROC-AUC | 0.9225 / 0.9886 |
| Coverage at $\tau$ / accuracy at $\tau$ | 84.7% / 100.00% |
| Selective accuracy @ 85% / 70% cov. | 100.00% / 100.00% |
| AURC | 0.0062 |
| Hate accuracy / macro-F1 | 96.50% / 0.9015 |

Table I summarises the results. At full coverage the model makes 0 false alarms and 77 missed fakes; every error is a

borderline claim. Under the calibrated threshold it answers 84.7% of claims without a single error.

### *B. Selective Prediction*

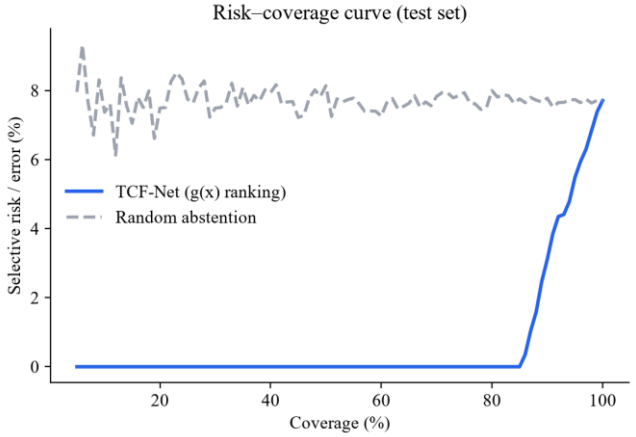



Fig. 2. Risk–coverage curve on the test set. Ranking by g(x) keeps the error at zero up to about 85% coverage, whereas random abstention stays near the full-coverage error.

Fig. 2 shows that error remains at zero until roughly 85% coverage and rises only when borderline claims are admitted, reaching 7.71% at full coverage; random abstention offers no benefit. The selection head assigns $g(x) \leq 0.20$ to every borderline test claim and $g(x) \geq 0.999$ to all others.

### *C. Analysis by Ambiguity and Language*

TABLE II
ACCURACY AND ABSTENTION BY SUBGROUP (TEST SET)

| **Subgroup** | **n** | **Acc.** | **Abstain** | **Hate acc.** |
|---|---|---|---|---|
| Borderline | 153 | 49.7% | 100.0% | 100.0% |
| Clear Fact | 423 | 100.0% | 0.0% | 100.0% |
| Sensational Deception | 423 | 100.0% | 0.0% | 91.7% |
| English | 350 | 91.7% | 14.6% | 90.0% |
| Hindi | 104 | 100.0% | 0.0% | 100.0% |
| Hinglish | 149 | 92.6% | 18.8% | 100.0% |
| Tamil | 190 | 91.0% | 17.4% | 100.0% |
| Tanglish | 206 | 90.3% | 19.9% | 100.0% |

Table II confirms that clear facts and sensational deceptions are classified perfectly in every language, while borderline claims sit at chance accuracy and are always deferred. Per-language accuracy differs only through the share of borderline items; Hindi has none and scores 100%. The hate head is precise but conservative, missing some hateful sensational posts.

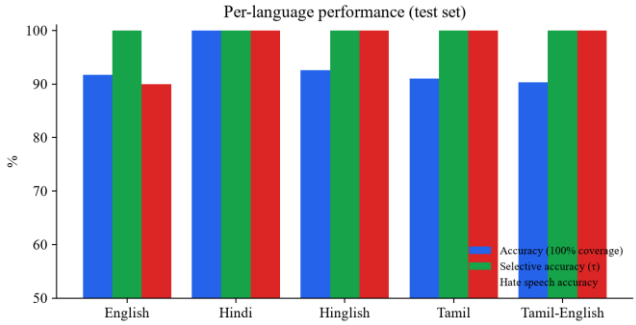



Fig. 3. Per-language accuracy at full coverage, selective accuracy at the calibrated threshold, and hate-speech accuracy on the test set.

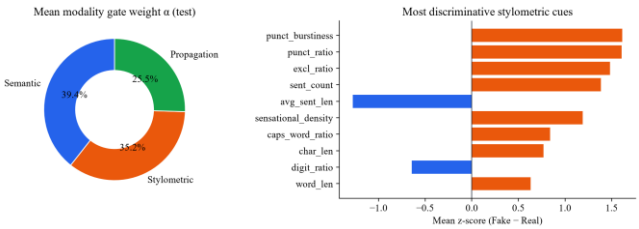



Fig. 4. Left: mean modality gate weights on the test set. Right: stylometric cues with the largest mean difference between fake and real claims.

### *D. Baselines and Ablation*

TABLE III
FULL-COVERAGE COMPARISON ON THE SAME SPLIT

| **Model** | **Acc.** | **Macro-F1** |
|---|---|---|
| TF-IDF (word) + Naive Bayes | 92.29% | 0.9225 |
| TF-IDF (word) + LR | 92.29% | 0.9225 |
| TF-IDF (word+char) + Linear SVM | 91.69% | 0.9167 |
| Stylometric (22-D) + LR | 91.59% | 0.9159 |
| MuRIL embeddings + LR | 91.59% | 0.9159 |
| TCF-Net (Semantic only) | 92.29% | 0.9225 |
| TCF-Net (Semantic + Stylometric) | 92.29% | 0.9225 |
| **TCF-Net (Full Tri-Modal, proposed)** | **92.29%** | **0.9225** |

All methods reach the same ≈92% ceiling at full coverage (Table III); differences amount to one or two test samples, so this benchmark cannot rank the modalities by accuracy. The practical advantage of TCF-Net is its reject option, which none of the baselines provides. The learned gate distributes weight across streams (semantic 39.4%, stylometric 35.2%, propagation 25.5%), and punctuation bursts, exclamations, capitals and sensational terms are the most discriminative stylometric cues.

### *E. Limitations*

The benchmark is template-generated, so absolute accuracy overstates performance on organic social-media text; evaluation on DFND, HinFakeNews and HASOC is the immediate next step. The propagation stream is a prior derived from text, not observed diffusion, and decisions are made without external evidence retrieval.

## VI. CONCLUSION

TCF-Net fuses semantic, stylometric and propagation evidence for multilingual and code-mixed credibility assessment, screens for hate speech in the same pass, and

abstains when the text does not support a verdict. On a five-language benchmark with leak-free evaluation it answers all verifiable claims correctly and defers every unverifiable one. Future work will fine-tune the encoder on real annotated corpora, replace the simulated cascade with observed sharing graphs, and use LLM-generated rationales as an additional advisory stream.

## REFERENCES


[1] S. Vosoughi, D. Roy, and S. Aral, "The spread of true and false news online," *Science*, vol. 359, no. 6380, pp. 1146–1151, 2018.

[2] H. T. Phan, N. T. Nguyen, and D. Hwang, "Fake news detection: A survey of graph neural network methods," *Appl. Soft Comput.*, vol. 139, Art. no. 110235, 2023.

[3] Y. Geifman and R. El-Yaniv, "SelectiveNet: A deep neural network with an integrated reject option," in *Proc. ICML*, PMLR vol. 97, 2019, pp. 2151–2159.

[4] A. Praseed, J. Rodrigues, and P. S. Thilagam, "Hindi fake news detection using transformer ensembles," *Eng. Appl. Artif. Intell.*, vol. 119, Art. no. 105731, 2023.

[5] E. Raja, B. Soni, S. K. Borgohain, and C. Lalrempuii, "Dravidian fake news detection with gradient accumulation based transformer model," in *Proc. ICON*, 2023, pp. 466–471.

[6] K. Sreelakshmi, B. Premjith, B. R. Chakravarthi, and K. P. Soman, "Detection of hate speech and offensive language CodeMix text in Dravidian languages using cost-sensitive learning approach," *IEEE Access*, vol. 12, 2024.

[7] T. Mandl *et al.*, "Overview of the HASOC subtrack at FIRE 2021: Hate speech and offensive content identification in English and Indo-Aryan languages and conversational hate speech," in *Proc. FIRE*, 2021.

[8] B. R. Chakravarthi *et al.*, "DravidianCodeMix: Sentiment analysis and offensive language identification dataset for Dravidian languages in code-mixed text," *Lang. Resour. Eval.*, vol. 56, pp. 765–806, 2022.

[9] S. Tufchi, A. Yadav, and T. Ahmed, "A comprehensive survey of multimodal fake news detection techniques: Advances, challenges, and opportunities," *Int. J. Multimedia Inf. Retr.*, vol. 12, 2023.

[10] B. Hu *et al.*, "Bad actor, good advisor: Exploring the role of large language models in fake news detection," in *Proc. AAAI*, vol. 38, no. 20, 2024, pp. 22105–22113.

[11] S. Khanuja *et al.*, "MuRIL: Multilingual representations for Indian languages," arXiv:2103.10730, 2021.

[12] W. O. Kermack and A. G. McKendrick, "A contribution to the mathematical theory of epidemics," *Proc. R. Soc. Lond. A*, vol. 115, no. 772, pp. 700–721, 1927.